

The Dimensional Speed-Limit Hypothesis

A One-Time-Dimension Framework for Dimension-Dependent Maximum Speed

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September 2026

Speculative theoretical hypothesis / toy model

Abstract

This paper develops a speculative kinematic framework in which the maximum speed C_n depends on the number N of physically accessible spatial dimensions, while time remains exactly one-dimensional. The observed 3+1 universe is the baseline, $C_3 = c$. A newly added spatial coordinate is written K_n . Its associated maximum velocity contribution is written k_n , so the central hypothesis is $C_n^2 = C_{n-1}^2 + k_n^2$. Therefore $C_n^2 = c^2 + \sum_{j=4}^n k_j^2$. An accessibility factor A_n , a toy wave model, and a compact fourth spatial dimension are introduced to explore how this recurrence could arise dynamically. If every fully accessible extra spatial dimension contributes $k_n = c$, then $C_n = c\sqrt{(N - 2)}$. The underlying Pythagorean geometry is standard mathematics; interpreting C_n as a dimension-dependent fundamental speed limit is the new speculative hypothesis.

1. Core Idea

The starting idea is simple. In ordinary geometry, adding a new independent spatial direction adds another squared component to the total magnitude. The proposed physical extension is that the maximum speed available in an N -dimensional spatial universe follows the same recursive pattern.

There is always one time coordinate t . The dimensional sequence is therefore 3+1, 4+1, 5+1, 6+1, Extra time dimensions are not introduced.

2. Symbols and Notation

N = number of spatial dimensions.

t = the single time coordinate.

c = the observed speed of light in our 3-spatial-dimensional universe.

C_n = proposed maximum speed when N spatial dimensions are accessible.

K_n = the newly added N -th spatial coordinate.

dK_n/dt = actual velocity of an object along the new spatial coordinate K_n .

k_n = maximum speed contribution associated with the new spatial dimension K_n .

A_n = accessibility of dimension K_n , ranging from 0 (inaccessible) to 1 (fully accessible).

R_n = characteristic size or compactification radius of the extra spatial dimension.

E_n = characteristic energy required to probe that extra dimension.

3. Central Speed-Limit Hypothesis

Our universe provides the baseline:

$$C_3 = c. \quad (1)$$

When the fourth spatial dimension K_4 becomes accessible:

$$C_4^2 = C_3^2 + k_4^2 = c^2 + k_4^2. \quad (2)$$

For a fifth spatial dimension:

$$C_5^2 = C_4^2 + k_5^2 = c^2 + k_4^2 + k_5^2. \quad (3)$$

The general recursive rule is therefore:

$$C_n^2 = C_{n-1}^2 + k_n^2. \quad (4)$$

Expanding the recursion:

$$C_n^2 = c^2 + \sum_{j=4}^n k_j^2. \quad (5)$$

Hence:

$$C_n = \sqrt{(c^2 + \sum_{j=4}^n k_j^2)}. \quad (6)$$

The coordinate K_n itself is a distance and is not directly added to a speed. The actual motion along that coordinate is dK_n/dt , with $|dK_n/dt| \leq k_n$.

4. Time Remains One-Dimensional

The model deliberately changes only the number of spatial dimensions. Time remains t at every dimensional level:

3D space: $(x, y, z ; t)$

4D space: $(x, y, z, K_4 ; t)$

5D space: $(x, y, z, K_4, K_5 ; t)$

N-D space: $(x_1, x_2, \dots, x_n ; t)$

Thus the proposal does not require imaginary time coordinates or additional time axes.

5. Spacetime Interval

A minimal flat-spacetime form for the proposal is:

$$ds_n^2 = dx_1^2 + dx_2^2 + \dots + dx_n^2 - C_n^2 dt^2. \quad (7)$$

For our 3+1 universe:

$$ds_3^2 = dx^2 + dy^2 + dz^2 - c^2 dt^2. \quad (8)$$

For four spatial dimensions:

$$ds_4^2 = dx^2 + dy^2 + dz^2 + dK_4^2 - C_4^2 dt^2. \quad (9)$$

The negative time term distinguishes time from the spatial coordinates. No imaginary coordinate is needed.

6. Dimensional Accessibility

An extra dimension can be mathematically present while remaining inaccessible to ordinary matter or energy. Let A_j measure how accessible the j -th extra spatial dimension is:

$$0 \leq A_j \leq 1.$$

Define a dimensionless coupling β_j and an effective factor:

$$\Gamma_n = 1 + \sum_{j=4}^n \beta_j A_j. \quad (10)$$

The proposed speed limit is:

$$C_n^2 = c^2 \Gamma_n. \quad (11)$$

If we define:

$$k_j^2 = c^2 \beta_j A_j, \quad (12)$$

then Eq. (11) immediately becomes:

$$C_n^2 = c^2 + \sum_{j=4}^n k_j^2, \quad (13)$$

which is the original recursive model.

7. Toy Wave Model

To show how C_n could appear as an actual propagation speed, consider a massless disturbance ϕ . A simple effective action is:

$$S_n = \frac{1}{2} \int dt d^N x [(1/c^2)(\partial\phi/\partial t)^2 - \Gamma_n |\nabla_n \phi|^2]. \quad (14)$$

Its wave equation is:

$$\partial^2 \phi / \partial t^2 - c^2 \Gamma_n \nabla_n^2 \phi = 0. \quad (15)$$

Since $C_n^2 = c^2 \Gamma_n$, this becomes:

$$\partial^2 \phi / \partial t^2 - C_n^2 \nabla_n^2 \phi = 0. \quad (16)$$

For a plane wave, the corresponding dispersion relation is:

$$\omega^2 = C_n^2 |q|^2. \quad (17)$$

Here q is the wave vector. In this toy model, C_n therefore acts as the propagation speed of the disturbance.

8. Concrete 4+1 Example

Let the fourth spatial coordinate be $K_4 = w$. Suppose it is compact, with characteristic radius R_4 . A compact spatial dimension naturally introduces a momentum scale $p_4 = \hbar/R_4$ and an energy scale of order:

$$E_4 = \hbar c/R_4. \quad (18)$$

This does not by itself imply a larger speed limit. It only supplies a physically meaningful scale at which the extra dimension could become observable.

As a toy activation law, take:

$$A_4(E) = 1 - \exp[-(E/E_4)^2]. \quad (19)$$

Let:

$$k_4^2 = \xi_4^2 c^2 A_4(E), \quad (20)$$

where ξ_4 controls the strength of the fourth-dimensional contribution. Then:

$$C_4(E) = c\sqrt{1 + \xi_4^2 A_4(E)}. \quad (21)$$

Using Eq. (19):

$$C_4(E) = c\sqrt{1 + \xi_4^2 [1 - \exp(-(E/E_4)^2)]}. \quad (22)$$

At low energy $E \ll E_4$, $A_4 \approx 0$ and $C_4 \approx c$. At high energy $E \gg E_4$, $A_4 \rightarrow 1$ and:

$$C_4 \rightarrow c\sqrt{1 + \xi_4^2}. \quad (23)$$

For the simple equal-contribution case $\xi_4 = 1$:

$$C_{4,\max} = \sqrt{2} c. \quad (24)$$

9. General N-Dimensional Case

For each additional dimension j , let:

$$E_j = \hbar c/R_j, \quad (25)$$

$$A_j(E) = 1 - \exp[-(E/E_j)^2], \quad (26)$$

$$k_j^2 = \xi_j^2 c^2 A_j(E). \quad (27)$$

Then:

$$C_n^2(E) = c^2 [1 + \sum_{j=4}^n \xi_j^2 A_j(E)]. \quad (28)$$

If every added spatial dimension is fully accessible and contributes equally, $\xi_j = 1$ and $A_j = 1$. Therefore:

$$C_n^2 = (N - 2)c^2, \quad (29)$$

and:

$$C_n = c\sqrt{N - 2}. \quad (30)$$

Examples are $C_3 = c$, $C_4 = \sqrt{2}c$, $C_5 = \sqrt{3}c$, and $C_6 = 2c$.

10. Causal Surface

For a null path, $ds_n^2 = 0$. Therefore:

$$(dx_1/dt)^2 + (dx_2/dt)^2 + \dots + (dx_n/dt)^2 = C_n^2. \quad (31)$$

The model therefore proposes two changes when an extra spatial dimension becomes accessible: a new orthogonal velocity component becomes available, and the maximum total velocity magnitude itself increases from C_{n-1} to C_n . The second statement is the non-standard part of the hypothesis.

11. Energy and Momentum

For a massless excitation in the toy wave model:

$$E^2 = C_n^2(p_1^2 + p_2^2 + \dots + p_n^2). \quad (32)$$

A possible massive extension would resemble:

$$E^2 = m^2 C_n^4 + C_n^2(p_1^2 + p_2^2 + \dots + p_n^2). \quad (33)$$

Equation (33) is only a provisional extension. If C_n itself depends on energy, the relation becomes self-consistent and implicit, so a complete theory must derive $E(p)$ rather than simply substitute $C_n(E)$ into the ordinary relativistic formula.

12. What Is Established and What Is New

Established mathematics and physics provide the use of orthogonal coordinates, N-dimensional norms, Lorentzian spacetime metrics, compact spatial dimensions, and Kaluza-Klein momentum scales.

The new hypothesis is specifically:

$$C_n^2 = C_{n-1}^2 + k_n^2.$$

Known extra-dimensional theories do not imply that adding a spatial dimension automatically increases the invariant speed of light. The accessibility function A_n , the coupling ξ_n , and the interpretation of C_n as a new causal limit are therefore proposed components of this model, not experimentally established facts.

13. Causality and Consistency

Using only one time dimension avoids the separate mathematical problems of multiple time coordinates, but it does not automatically make faster-than-c propagation paradox-free. If a higher-dimensional signal appears faster than c to observers restricted to 3+1 dimensions, its causal interpretation depends on the transformation laws of the complete theory.

A successful version of the model must therefore derive a consistent symmetry structure, explain how higher-dimensional and ordinary observers transform between frames, recover ordinary special relativity when extra dimensions are inaccessible, and show that no forbidden causal loops are produced.

14. Experimental Tests

A physical realization could potentially produce modified particle dispersion, missing momentum into extra dimensions, threshold-dependent propagation, short-distance changes in gravity, or Lorentz-violating effects. Existing collider experiments, astrophysical timing measurements, precision Lorentz tests, and short-range gravity experiments place strong constraints on such effects.

The hypothesis becomes scientifically useful when the parameters R_n , ξ_n , and A_n are derived or bounded strongly enough to make numerical predictions.

15. Conclusion

The Dimensional Speed-Limit Hypothesis proposes that our familiar speed limit c is C_3 , the limiting speed associated with three accessible spatial dimensions. Each newly accessible spatial dimension K_n contributes a new maximum velocity component k_n according to:

$$C_n^2 = C_{n-1}^2 + k_n^2.$$

Time remains one-dimensional throughout the hierarchy. The model can be written using the spacetime interval $ds_n^2 = \sum dx_i^2 - C_n^2 dt^2$ and represented by a toy wave theory in which dimensional accessibility changes the propagation coefficient. Compact extra dimensions provide a natural activation scale $E_n = \hbar c/R_n$. Under equal fully activated contributions, the model predicts $C_n = c\sqrt{N-2}$.

The recursion is mathematically clear, but its interpretation as a fundamental law of nature remains speculative. The decisive next step is to derive k_n from a deeper geometry or field dynamics and test the resulting predictions against established relativistic and experimental constraints.

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